Mathieu Helfter

Postdoctoral Researcher at ISTA

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Research interests: dynamical systems and fractal geometry

Education

2021–2024 **PhD in Mathematics**, *Sorbonne Université, IMJ-PRG, CNRS*,
Thesis: "Complexity and Bifurcation of the Identity."
Supervisor: Pierre Berger

2020–2021 **Internship with Dmitry Turaev and Pierre Berger**

2019–2020 **Master's Degree in Random Models and Probability**, *LPSM, Sorbonne Université*, Highest Honors

2018–2019 **M1 Jacques Hadamard – Mathematics and Applications Track**, *École Normale Supérieure (ENS) Paris-Saclay*

2017–2018 **Bachelor's Degree (L3) in Mathematics**, *ENS Paris-Saclay*

2014–2017 **Preparatory Classes (MP*)**, *Lycée Louis-le-Grand*

2014 **French Baccalauréat (Scientific Track)**, Highest Honors

Positions

Oct. 2024–present **Postdoctoral Researcher**, *Institute of Science and Technology Austria (ISTA)*,
Kaloshin Group; supervisor: Vadim Kaloshin

Publications and Preprints

1. **Every Diffeomorphism Is a Total Renormalization of a Map Close to the Identity**, *2024*, with P. Berger and N. Gourmelon; Inventiones Mathematicae, `https://doi.org/10.1007/s00222-024-01305-w`
2. **Scales**, *Mathematische Zeitschrift*, 2025, `https://doi.org/10.1007/s00209-025-03719-5`
3. **Sets with Arbitrary Hausdorff and Packing Scales in Infinite-Dimensional Banach Spaces**, *2025*, Journal of Fractal Geometry, `https://ems.press/journals/jfg/articles/14299261`
4. **Infinite Dimensional Multifractal Analysis of the Wiener Measure**, *with Ai-Hua Fan*, `https://arxiv.org/abs/2512.06984`
5. **Jet rigidity of Mather's β-function and complexified KAM curves holomorphic in potential for analytic standard maps**, *2026*, `https://arxiv.org/abs/2609.15617`

Other Scientific Contributions

Sep. 2025 **Near a Stagnant Dynamical System, with Time and a Magnifying Glass, Anything Can Happen**, *Scientific outreach article*, Lettres de l'INSMI; with Pierre Berger and Nicolas Gourmelon, (in french)

https://www.insmi.cnrs.fr/fr/cnrsinfo/pres-dune-dynamique-stagnante-avec-le-temps-et-une-loupe-tout-peut-arriver

Oct. 2024 **Complexity and Bifurcation of the Identity**, *PhD thesis manuscript*, under the supervision of Pierre Berger, `https://theses.hal.science/tel-04906262/document`

Conferences and Seminars – Talks Given

14–18 Sep. 2026 **Fields–CNAM Nonlinear Days 2026**, *Fields Institute*, Toronto, Canada

17 Jun. 2026 **New Directions in Thermodynamic Formalism**, *CIRM*, Marseille, France

8 Jun. 2026 **Fractals and Related Fields 5**, *Porquerolles*, France

18–22 May 2026 **Real and Complex Wild Dynamics**, *UiT The Arctic University of Norway*, Tromsø, Norway

17 Apr. 2026 **Budapest–Vienna Dynamics Seminar (BudWiser)**

8 Apr. 2026 **Internal Dynamics Seminar**, *ISTA*, Austria

4 Feb. 2026 **Séminaire de l'équipe SD**, *Université Paris XIII*, France

13 Jan. 2026 **MathPhys Analysis Seminar**, *ISTA*, Austria

6 Jan. 2026 **Wild Dynamics Working Group**, *IMJ-PRG, Sorbonne Université*, France

5 Jan. 2026 **Ergodic Theory and Dynamical Systems Working Group**, *Laboratory of Mathematics of Orsay*, France

17 Oct. 2025 **BudWiser**, *ISTA*, Klosterneuburg, Austria

10 Apr. 2025 **Seminar at Wuhan University**, *Wuhan University*, China

17 Dec. 2024 **Wild Dynamical Systems Working Group**, *IMJ-PRG, Sorbonne Université*, France

13 Dec. 2024 **Dynamical Systems Seminar**, *IMJ-PRG, Sorbonne Université*, France

13–17 Nov. 2024 **Workshop on Dynamical Systems and Related Topics**, *Penn State University*, United States

7 Nov. 2024 **Maryland Dynamics Seminar**, *University of Maryland*, United States

7 Oct. 2024 **PhD Defence**, *IMJ-PRG, Sorbonne Université*, France

10–14 Jun. 2024 **Equadiff 2024**, *Karlstad University*, Sweden

27 Feb. 2024 **DynamIST Seminar**, *ISTA*, Austria

23 Feb. 2024 **BudWiser**, *Budapest University of Technology and Economics*, Budapest, Hungary

27 Nov.–1 Dec. 2023 **Lyapunov Exponents and Smooth Ergodic Theory**, *Paraty-RJ*, Brazil

26–31 Mar. 2023 **French–Japanese Workshop on Real and Complex Hénon Maps**, *Kyoto University*, Japan

12–20 Mar. 2022 **Focused Research Group: Computability and Complexity of Statistical Behaviour of Dynamical Systems**, *BIRS, Banff*, Canada

6–10 Sep. 2021 2nd **Workshop on Wild Dynamical Systems**, *São Miguel, Azores*, Portugal

Conference and Seminar Organization

Sep. 2026 **DynamIST Summer School 2026**, *Organizer*, Strobl, Austria

Sep. 2025 **DynamIST Summer School 2025**, *Organizer*, Strobl, Austria

2024–present **DynamIST Weekly Seminar**, *Co-organizer*, ISTA, Klosterneuburg, Austria

2022–2024 **Weekly Working Group on Wild Dynamical Systems**, *Co-organizer*, IMJ-PRG, Sorbonne Université

11 Sep.–15 Dec. 2023 **French–Japanese Workshop on Real and Complex Dynamics**, *Local organizer*, IMJ-PRG, Sorbonne Université

Teaching

Autumn 2026 Co-lecturer and course developer for dynamical systems for PhD students in other disciplines, with Alessandra Nardi (ISTA)

Spring 2021 Exercise-session instructor for first-year undergraduate mathematics, Jussieu

Research Internships

Nov. 2020–Jun. 2021 **Renormalization of Analytic Maps Close to the Identity**, *Supervisor: D. Turaev*, Imperial College London (remote)

Apr.–Aug. 2019 **Study of Generative Neural Networks (VAE, β-VAE, GANs, InfoGANs)**, *Supervisors: B. Taetz and D. Stricker*, Technical University of Kaiserslautern, Germany

Feb.–Jul. 2018 **Modeling Body Movements Using Graph Signals**, *Supervisor: A. Kalogeratos*, Centre for Mathematics and Their Applications, Cachan, France

Languages

French Native language

English Cambridge Advanced English (C1 level)

Computer Skills

Office Software Advanced proficiency

LaTeX Advanced proficiency

Python Basic knowledge

Shadertoy Basic knowledge

Interests

Sports Climbing, hiking, cycling

Arts Music production, percussion